Aftertalk

FINAL SUBMISSION · APRIL 30, 2026

github.com/theaayushstha1/aftertalk

AFTERTALK · V0.9.0-RC1

ALL REQUIRED + 5 OF 5 STRETCHES · ZERO NETWORK REQUESTS · 45 OF 45 TESTS PASSING

Fully on-device meeting intelligence for iOS 26, with a verifiable *privacy posture*.

Aftertalk is a complete iPhone application that records a meeting, transcribes it locally, generates a structured summary through Apple Foundation Models, and answers spoken or typed questions about the recording, both within a single meeting and across the entire library. The recording, transcription, summarization, retrieval, generation, and synthesis pipelines all execute on the device. The repository contains a static audit demonstrating that no production code path imports a networking primitive, and a runtime monitor raises a fault if any active network interface is detected during recording. The screen recorded demonstration accompanies this submission with airplane mode visible in the iOS Control Center for the full duration of the run.

System architecture

CAPTURE, SUMMARIZE, RETRIEVE, ANSWER

Audio is captured through **AVAudioEngine** at 48 kHz, downsampled to 16 kHz, and routed through an energy based voice activity gate before reaching **Moonshine small streaming** for real time transcript deltas. On stop, **FluidAudio Parakeet TDT 0.6B v2** produces a polished canonical transcript with word level timings, while **Pyannote 3.1 with WeSpeaker v2** performs speaker diarization concurrently. The reconciled transcript is chunked and embedded with **Apple NLContextualEmbedding** (512 dimensional, zero bundle bytes) into a typed **SwiftDataVectorStore**. **Apple Foundation Models** emits the structured summary through the `@Generable` macro, covering decisions, action items with attributable owners, topics, and open questions. Question answering routes through a hybrid retriever combining **BM25** with dense cosine similarity under **Reciprocal Rank Fusion**; short transcripts skip retrieval and present the full text under the 4096 token budget. Responses are synthesized by **FluidAudio Kokoro 82M** on the Neural Engine, dispatched sentence by sentence as the model streams.

Decisions of consequence

DECISION

Two tier ASR over a single model

Live preview uses **Moonshine small streaming** because it sustains real time on long readings; sustained tests with the medium variant exhibited multi minute latency accumulation. The polished canonical transcript is produced by **Parakeet** on the saved waveform after recording terminates. The two tier design preserves time to first token for the user facing surface while delivering Whisper class accuracy in the stored artifact.

DECISION

Foundation Models over MLX with a bundled weight

Foundation Models is system provided on iOS 26 and ships with zero bytes of language model weight in the bundle, in contrast to MLX with Phi 4 mini at approximately 2 GB. The `@Generable` macro produces typed structured summaries without a JSON parsing layer. Throughput on A18 hardware sits near 30 tokens per second, and the 4096 token budget surfaces retrieval failures during development rather than in front of users.

DECISION

Hybrid retrieval with full transcript shortcut

Short meetings under approximately 10 000 characters bypass retrieval and present the full transcript to the language model. Longer meetings invoke a hybrid pipeline combining BM25 with dense cosine similarity through Reciprocal Rank Fusion at constant k equals 60. A grounding gate refuses to invoke generation when retrieval recovers no chunks above threshold and no summary fallback exists, returning a fixed disclaimer instead.

DECISION

Verifiable privacy over claimed privacy

A static repository audit demonstrates that `URLSession`, `URLRequest`, and arbitrary loads exemptions are absent from the production Swift sources. A runtime `NWPathMonitor` assertion raises a fault when any active network interface is detected during recording. Pasteboard exports are local only with a ten minute expiration; recordings and performance logs are stamped with iOS file protection at the first user authentication level.

Verification

0	45/45	5/5	2.3GB	82M
NETWORK REQUESTS	UNIT TESTS PASSING	STRETCH GOALS SHIPPED	PEAK MEMORY, ON DEVICE	KOKORO PARAMS ON ANE

SUBMISSION · ACCOMPANYING MATERIAL

The repository is available publicly at `github.com/theaayushstha1/aftertalk` under the MIT license, tagged at `v0.9.0-rc1`. The full architectural narrative, fifteen documented decisions, and complete dependency provenance are present in the repository `README.md`, `ARCHITECTURE.md`, `DECISIONS.md`, and `docs/Aftertalk-Complete-Reference.pdf`. **A screen recorded demonstration video, captured on iPhone 17 Pro Max with airplane mode visible throughout, will be attached to this email by tonight** and shows the complete loop: airplane mode confirmation, recording, structured summary, per meeting question answering, cross meeting question answering, and the in app privacy audit.